

Linear constraints

Active constraints

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Optimization: principles and algorithms



ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

Active constraints

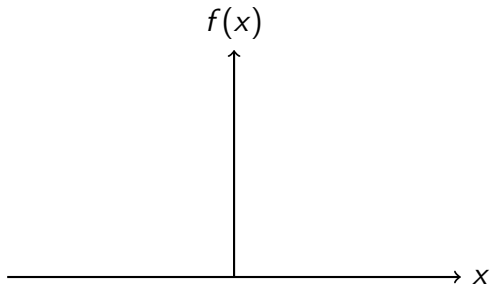
Active constraints are somehow constraints that matter. We will see that, in linear optimization, finding the optimum solution amounts to finding the constraints that are active at the solution.

First example

$$\min_{x \in \mathbb{R}} x^2$$

subject to

$$-10 \leq x \leq 4.$$

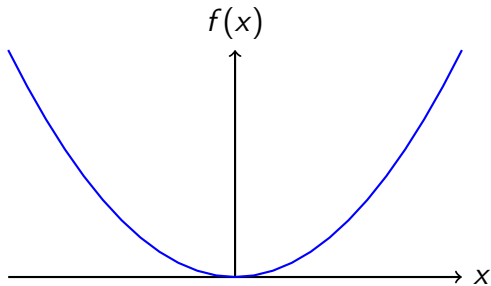


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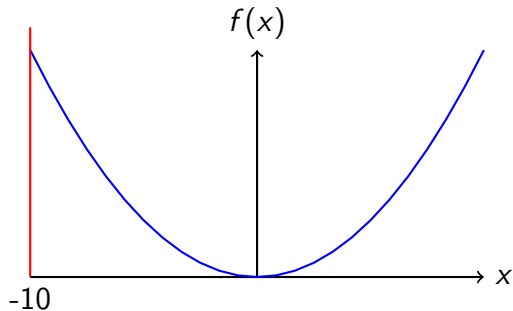


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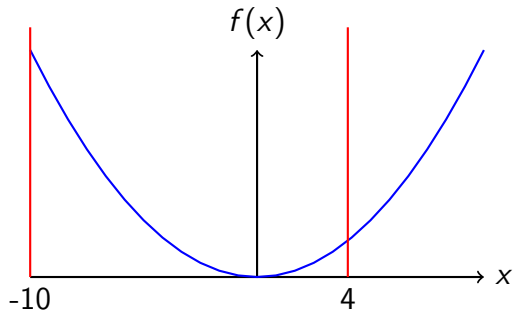


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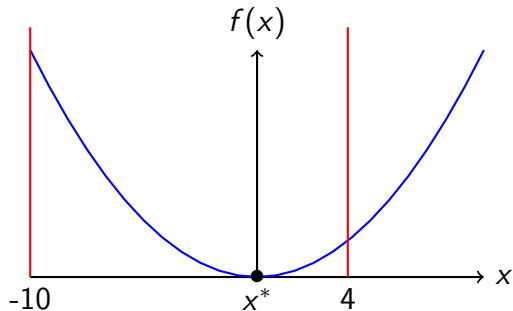


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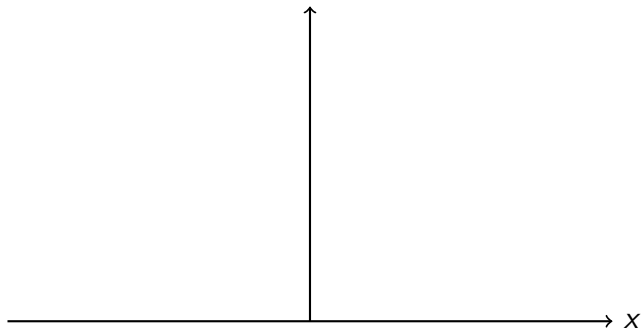
Second example

subject to

$$\min_{x \in \mathbb{R}} x^2$$

$$1 \leq x \leq 4.$$

$$f(x)$$



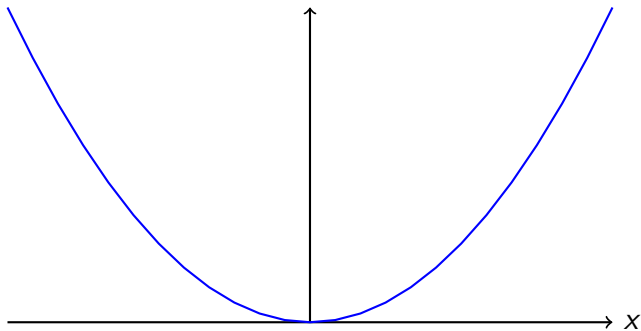
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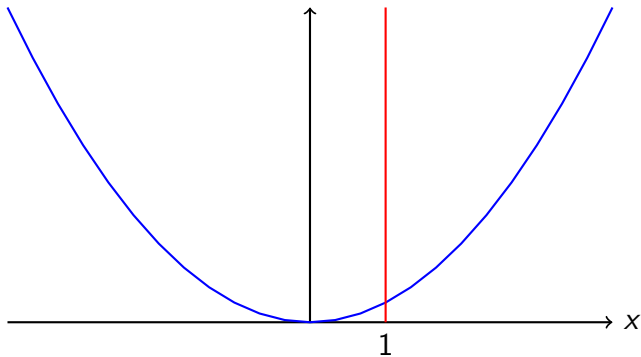
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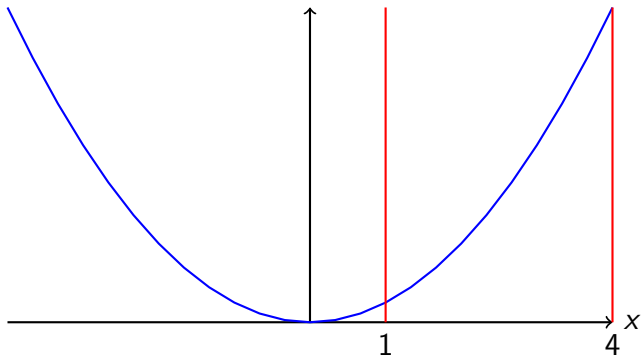
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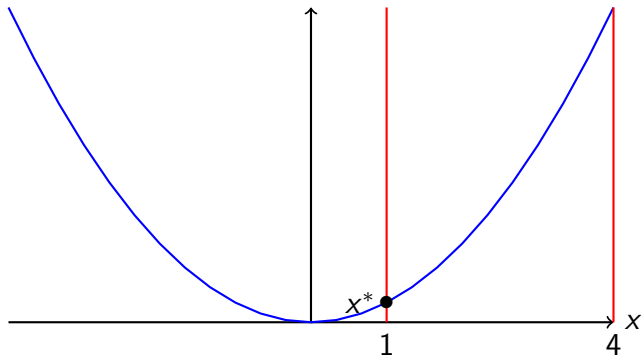
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Definition

Inequality constraints

Consider $g : \mathbb{R}^n \rightarrow \mathbb{R}$ and the constraint

$$g(x) \leq 0.$$

It is active at x^* if

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Equality constraints

Consider $h : \mathbb{R}^n \rightarrow \mathbb{R}$ and the constraint

$$h(x) = 0.$$

It is active at x^* if

$$h(x^*) = 0.$$

Optimization

- Consider the problem

$$\min_{x \in \mathbb{R}^n} f(x)$$

subject to

$$g(x) \leq 0,$$

where $g : \mathbb{R}^n \rightarrow \mathbb{R}^p$.

Optimization

- Consider the problem

$$\min_{x \in \mathbb{R}^n} f(x)$$

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$$g(x) \leq 0,$$

where $g : \mathbb{R}^n \rightarrow \mathbb{R}^p$.

- x^* is a local optimum and

$$\mathcal{A}(x^*) = \{i = 1, \dots, p \mid g_i(x^*) = 0\}.$$

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where $g : \mathbb{R}^n \rightarrow \mathbb{R}^p$.

- x^* is a local optimum and

$$\mathcal{A}(x^*) = \{i = 1, \dots, p \mid g_i(x^*) = 0\}.$$

- x^* is also a local optimum of

$$\min_{x \in \mathbb{R}^n} f(x)$$

subject to

$$g_i(x) = 0, \forall i \in \mathcal{A}(x^*).$$

Summary

Active constraints are constraints that are verified with equality.

A constraint is active at x if x is “on the constraint”.

If we happen to know the inequality constraints that are active at a solution, we can drop inactive constraints, and replace inequality constraints by equality.

This is useful only at a theoretical level, as it requires the knowledge of x^ .*